

Online Retail Dataset Analysis Report

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This project focused on the cleaning, exploration, and analysis of an online retail transactions dataset using Python and Power BI. The objective was to identify and resolve data quality issues, structure the dataset for reliable analysis, and generate meaningful insights through exploratory data analysis to simulate a real-world business scenario.

The raw dataset contained 541,910 transaction records and included key variables such as Invoice Number, Stock Code, Description, Quantity, Invoice Date, Unit Price, Customer ID, Country, and Total Sales. During the familiarization stage, the dataset structure and data types were reviewed to identify numerical, categorical, and date-based variables. Quantity, Unit Price, and Total Sales were classified as numerical features, while Invoice Number, Stock Code, Description, Customer ID, and Country were treated as categorical variables. InvoiceDate was converted into a datetime format to support time-series analysis.

Several data cleaning procedures were carried out to improve data quality and consistency. Missing values in the Description and CustomerID columns were identified and handled appropriately. Duplicate records were removed to maintain transactional integrity, while invalid transactions such as negative quantities and zero or negative unit prices were excluded from the analysis. Text fields were standardized for consistency, and country entries were validated to ensure uniform formatting. Non-product stock codes, including POST, DOT, and BANK CHARGES, were removed because they represented operational charges rather than retail products.

Exploratory Data Analysis (EDA) was conducted using summary statistics and visualizations to identify patterns and trends within the dataset. Median values for Quantity, UnitPrice, and TotalSales were calculated to better understand transaction behaviour. Additional analysis included total revenue, average order value, top-performing products, and high-value customers.

Bar charts and time-series plots were created to visualize trends and communicate key findings. A bar chart identified the top 10 products by revenue, while a daily sales trend chart examined revenue changes over time. Country-level analysis was also conducted to identify the highest revenue-generating countries.

Key business metrics analyzed included Total Revenue, Total Orders, and Total Customers. The dataset generated approximately 8.74 million in revenue from over 18,000 orders and more than 4,000 customers, showing that a relatively small number of products and customers contributed a large share of total revenue.

This project was completed using Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn, Jupyter Notebook for workflow documentation, and Power BI for dashboard development and visualization.